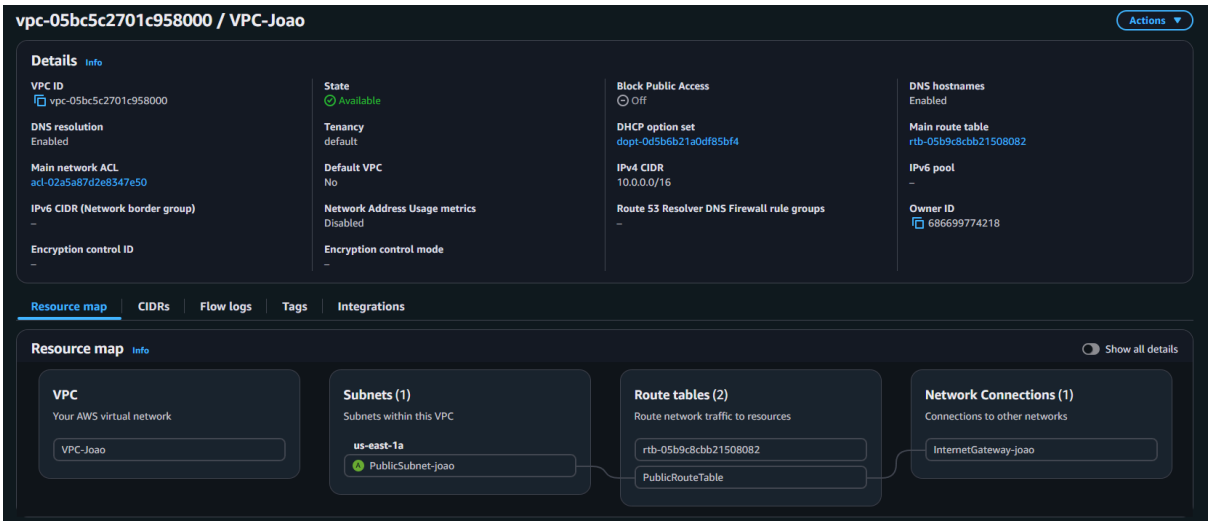
João-VPC:



Terraform apply output:

```
joaoribeiro@JOCAPC:~/terraform-vpc-lab$ terraform apply -var="key_pair_name=Joao-useast1" --auto-approve
data.aws_ami.amazon_linux: Reading...
aws_vpc.my_vpc: Refreshing state... [id=vpc-05bc5c2701c958000]
data.aws_ami.amazon_linux: Read complete after 1s [id=ami-0e10497160c48e829]
aws_internet_gateway.igw: Refreshing state... [id=igw-0c1ec4da444164b8d]
aws_subnet.public_subnet: Refreshing state... [id=subnet-0fcb490b50b61a1c9]
aws_security_group.ec2_sg: Refreshing state... [id=sg-0b6befd5fb6c1c1df]
aws_route_table.public_rt: Refreshing state... [id=rtb-02827d48d0d02b130]
aws_route_table_association.public_subnet_assoc: Refreshing state... [id=rtbassoc-06157649fb0fb79f2]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the
following symbols:
  + create

Terraform will perform the following actions:

  # aws_instance.web_server will be created
  + resource "aws_instance" "web_server" {
      + ami                                  = "ami-0e10497160c48e829"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = (known after apply)
      + availability_zone                    = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + force_destroy                        = false
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_lifecycle                   = (known after apply)
      + instance_state                       = (known after apply)
      + instance_type                        = "t2.micro"
      + ipv6_address_count                   = (known after apply)
      + ipv6_addresses                       = (known after apply)
      + key_name                             = "Joao-useast1"
      + monitoring                           = (known after apply)
      + outpost_arn                          = (known after apply)
      + password_data                        = (known after apply)
      + placement_group                      = (known after apply)
      + placement_group_id                   = (known after apply)
      + placement_partition_number           = (known after apply)
      + primary_network_interface_id         = (known after apply)
      + private_dns                          = (known after apply)
      + private_ip                           = (known after apply)
      + public_dns                           = (known after apply)
      + public_ip                            = (known after apply)
      + region                               = "us-east-1"
      + secondary_private_ips                = (known after apply)
      + security_groups                      = (known after apply)
```

```
      + source_dest_check                    = true
      + spot_instance_request_id             = (known after apply)
      + subnet_id                            = "subnet-0fcb490b50b61a1c9"
      + tags                                 = {
          + "Name" = "WebServerInstance-joao"
        }
      + tags_all                             = {
          + "Name" = "WebServerInstance-joao"
        }
      + tenancy                              = (known after apply)
      + user_data_base64                     = (known after apply)
      + user_data_replace_on_change          = false
      + vpc_security_group_ids               = [
          + "sg-0b6befd5fb6c1c1df",
        ]

      + capacity_reservation_specification (known after apply)

      + cpu_options (known after apply)

      + ebs_block_device (known after apply)

      + enclave_options (known after apply)

      + ephemeral_block_device (known after apply)

      + instance_market_options (known after apply)

      + maintenance_options (known after apply)

      + metadata_options (known after apply)

      + network_interface (known after apply)

      + primary_network_interface (known after apply)

      + private_dns_name_options (known after apply)

      + root_block_device (known after apply)

      + secondary_network_interface (known after apply)
    }

Plan: 1 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + instance_public_ip = (known after apply)
aws_instance.web_server: Creating...
aws_instance.web_server: Still creating... [00m10s elapsed]
aws_instance.web_server: Still creating... [00m20s elapsed]
aws_instance.web_server: Still creating... [00m30s elapsed]
aws_instance.web_server: Creation complete after 34s [id=i-022602247162fdeca]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
```

```
Outputs:

instance_public_ip = "100.26.150.130"
joaoribeiro@JOCAPC:~/terraform-vpc-lab$
```

SSH connection to EC2 instance:

```
joaoribeiro@JOCAPC:~$ ssh -i "Joao-useast1.pem" ec2-user@ec2-100-26-150-130.compute-1.amazon
aws.com
The authenticity of host 'ec2-100-26-150-130.compute-1.amazonaws.com (100.26.150.130)' can't
 be established.
ED25519 key fingerprint is SHA256:LUnfuYC1DNs+FJ0/J8C9wdt8DQdN9ftjSuMyK62LyxA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-100-26-150-130.compute-1.amazonaws.com' (ED25519) to the lis
t of known hosts.
       #_
       ~\_  ####_        Amazon Linux 2
      ~~  \_#####\
      ~~     \###|       AL2 End of Life is 2026-06-30.
      ~~       \#/ ___
       ~~       V~' '->
        ~~~         /    A newer version of Amazon Linux is available!
          ~~._.   _/
             _/ _/       Amazon Linux 2023, GA and supported until 2028-03-15.
           _/m/'           https://aws.amazon.com/linux/amazon-linux-2023/

[ec2-user@ip-10-0-1-31 ~]$
```

Running instance:

Instance summary for i-022602247162fdeca (WebServerInstance-joao) Info

Connect | Instance state | Actions

Updated less than a minute ago

Instance ID i-022602247162fdeca	**Public IPv4 address** 100.26.150.130 | open address	**Private IPv4 addresses** 10.0.1.31
IPv6 address –	**Instance state** Running	**Public DNS** ec2-100-26-150-130.compute-1.amazonaws.com | open address
Hostname type IP name: ip-10-0-1-31.ec2.internal	**Private IP DNS name (IPv4 only)** ip-10-0-1-31.ec2.internal	
Answer private resource DNS name –	**Instance type** t2.micro	**Elastic IP addresses** –
Auto-assigned IP address 100.26.150.130 [Public IP]	**VPC ID** vpc-05bc5c2701c958000 (VPC-Joao)	**AWS Compute Optimizer finding** Opt-in to AWS Compute Optimizer for recommendations. | Learn more
IAM role –	**Subnet ID** subnet-0fcb490b50b61a1c9 (PublicSubnet-joao)	**Auto Scaling Group name** –
IMDSv2 Optional EC2 recommends setting IMDSv2 to required | Learn more	**Instance ARN** arn:aws:ec2:us-east-1:686699774218:instance/i-022602247162fdeca	**Managed** false
Operator –		